

Solution 13

Exercise 1 (Exam 2018) [2pt. | 2pt. | 4pt.]

Consider the system

$$x(k+1) = Ax(k) + Bu(k)$$

and the standard MPC control law $\kappa(x(k)) = u_0^*(x(k))$ as defined in the supplementary material by optimization problem $P_{S,1}$, satisfying Assumptions S.1-1, S.1-2, S.1-3, with polytopic $\mathcal{X}$, $\mathcal{U}$, $\mathcal{X}_f$ and quadratic stage and terminal cost.

i) Mark the true statements about optimization problem $P_{S,1}$.

	True	False
$P_{S,1}$ is a convex optimization problem.	☐	☐
$P_{S,1}$ is a linear program.	☐	☐
$P_{S,1}$ has a bounded optimal cost.	☐	☐
$P_{S,1}$ has a unique minimizer.	☐	☐

ii) Mark the true statements about the closed-loop system under the MPC control law $\kappa(x(k))$. Assume that $P_{S,1}$ is feasible for $x_0 = x(0)$.

	True	False
$P_{S,1}$ is feasible for all $k \geq 0$.	☐	☐
$x(k) \in \mathcal{X}$ and $u(k) \in \mathcal{U}$ for all $k \geq 0$.	☐	☐
$x(N) \in \mathcal{X}_f$.	☐	☐
$x(k+1)^T Q x(k+1) \leq x(k)^T Q x(k)$.	☐	☐

iii) Show that the infinite-horizon cost of the closed-loop system is always smaller or equal to the predicted open-loop cost, i.e.

$$J_\infty(x(0)) = \sum_{k=0}^{\infty} \|x(k)\|_Q^2 + \|u(k)\|_R^2 \leq J^*(x(0)),$$

for $u(k) = \kappa(x(k))$.

Hint: Remember that in the standard MPC formulation under Assumptions S.1-1, S.1-2, S.1-3, it holds that $J^*(x(k+1)) - J^*(x(k)) \leq -\|x(k)\|_Q^2 - \|u(k)\|_R^2$.

Solution:

- i) True - convex quadratic cost, linear dynamics, polytopic constraints
False - quadratic cost
True - convex problem with strictly convex cost
True - convex problem with strictly convex cost

- ii) True - MPC with invariant terminal set is recursively feasible
 True - since feasible
 False - open-loop prediction $\neq$ closed loop trajectory (example with terminal set 0 and short horizon should make this intuitively clear)
 False - closed-loop state does not need to decrease in Q-norm at every time step
- iii) From the hint we have that $\|x(k)\|_Q^2 + \|u(k)\|_R^2 \leq J^*(x(k)) - J^*(x(k+1))$, i.e., it holds that

$$\begin{aligned} J_\infty(x(0)) &= \sum_{k=0}^{\infty} \|x(k)\|_Q^2 + \|u(k)\|_R^2 \\ &\leq \sum_{k=0}^{\infty} J^*(x(k)) - J^*(x(k+1)). \end{aligned}$$

Expanding the summation leads to “telescopic sum”:

$$\begin{aligned} \sum_{k=0}^{\infty} J^*(x(k)) - J^*(x(k+1)) &= J^*(x(0)) - J^*(x(1)) + J^*(x(1)) - J^*(x(2)) + \dots \\ &= \lim_{\bar{k} \rightarrow \infty} J^*(x(0)) - J^*(x(\bar{k})). \end{aligned}$$

Due to asymptotic (specifically exponential) stability we finally have

$$\lim_{\bar{k} \rightarrow \infty} -J^*(x(\bar{k})) = 0$$

which establishes

$$J_\infty \leq \sum_{k=0}^{\infty} J^*(x(k)) - J^*(x(k+1)) = J^*(x(0)).$$

Exercise 2 (Exam 2020) [2pt. | 3pt. | 2pt. | 1pt. | 3pt. | 2pt. | 4pt.]

Consider the following scalar system

$$x(k+1) = 2x(k) + u(k).$$

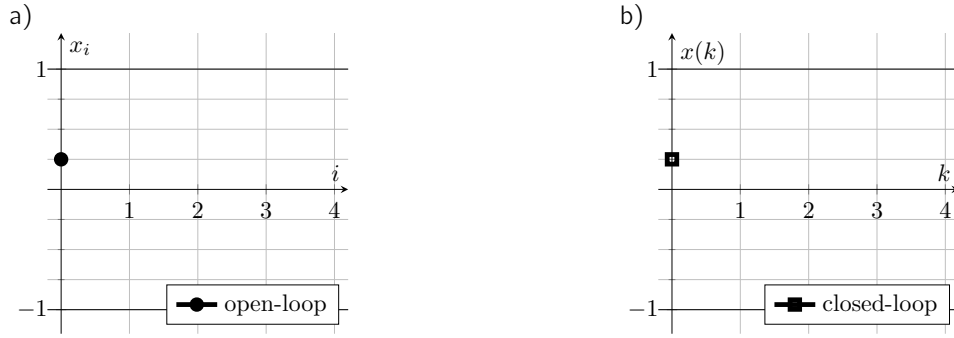
The system is subject to state and input constraints:

$$\begin{aligned} -1 &\leq x \leq 1, \\ -\frac{1}{3} &\leq u. \end{aligned}$$

The goal is to keep the system within the constraints while minimizing the input to the system. A simple MPC control law is proposed

$$\begin{aligned} J_{N,1}^*(x(k)) &= \underset{X, U}{\text{minimize}} \quad \sum_{i=0}^{N-1} u_i^2 \\ \text{subject to} \quad &x_{i+1} = 2x_i + u_i, \quad i = 0, \dots, N-1, \\ &|x_i| \leq 1, \quad i = 0, \dots, N, \\ &u_i \geq -\frac{1}{3}, \quad i = 0, \dots, N-1, \\ &x_0 = x(k), \end{aligned} \tag{1}$$

where $N = 2$ and the control law is $\kappa(x(k)) = u_0^*(x(k))$.



- For the initial state $x(0) = 0.25$, provide the optimal open-loop prediction resulting from the MPC problem (1) and plot it in Figure a).
- State the closed-loop system evolution under the MPC controller (1) for two time steps, $\{x(1), x(2)\}$ and plot it in Figure b).
- Provide the feasible set of the MPC problem (1) of the form $\mathcal{X}_{N=2}^{(1)} = \{x | H_N x \leq h_N\}$.
- Is the MPC problem (1) guaranteed to be recursively feasible for any initially feasible state? Justify your answer.

Consider now the modified MPC problem with added terminal constraint $x_N = 0$.

$$\begin{aligned}
 J_{N,2}^*(x(k)) = & \underset{X, U}{\text{minimize}} \quad \sum_{i=0}^{N-1} u_i^2 \\
 \text{subject to} \quad & x_{i+1} = 2x_i + u_i, \quad i = 0, \dots, N-1, \\
 & |x_i| \leq 1, \quad i = 0, \dots, N, \\
 & u_i \geq -\frac{1}{3}, \quad i = 0, \dots, N-1, \\
 & x_N = 0, \\
 & x_0 = x(k),
 \end{aligned} \tag{2}$$

where $N = 2$.

- Provide the feasible set of the MPC problem (2) with horizon $N = 2$ of the form $\mathcal{X}_{N=2}^{(2)} = \{x | H_N x \leq h_N\}$.
- Is the MPC problem (2) with horizon $N = 2$ guaranteed to be recursively feasible? Justify your answer.
- Mark all statements that are guaranteed to be true.

	True
The feasible set of the MPC problem (1) with planning horizon $N=2$ is a subset of the feasible set of the MPC problem (1) with planning horizon $N=3$, i.e., $\mathcal{X}_{N=2}^{(1)} \subseteq \mathcal{X}_{N=3}^{(1)}$.	☐
The feasible set of the MPC problem (2) with planning horizon $N=2$ is a subset of the feasible set of the MPC problem (2) with planning horizon $N=3$, i.e., $\mathcal{X}_{N=2}^{(2)} \subseteq \mathcal{X}_{N=3}^{(2)}$.	☐
For fixed horizon N , the optimal open-loop cost $J_{N,2}^*(x(k))$ of the MPC problem (2) is smaller or equal to the optimal open-loop cost $J_{N,1}^*(x(k))$ of the MPC problem (1) for all $x(k)$ for which both optimization problems are feasible.	☐
For the MPC problem (2) with two different prediction horizons N_1 and N_2 with $N_1 > N_2$ the optimal open-loop cost $J_{N_1,2}^*(x(k))$ is larger or equal to $J_{N_2,2}^*(x(k))$, i.e. $J_{N_1,2}^*(x(k)) \geq J_{N_2,2}^*(x(k))$.	☐

Solution:

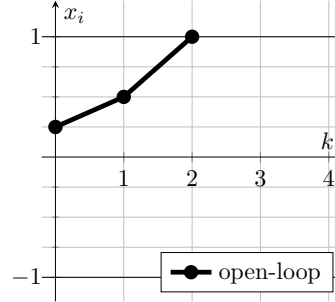
- i) Notice cost is minimized by choosing input $u_0 = u_1 = 0$.

$$x_0 = x(0) = 0.25$$

$$x_1 = 2x_0 + 0 = 0.5$$

$$x_2 = 2x_1 + 0 = 1$$

Plot open-loop prediction in Figure a)



- ii) Obtain $x(1) = 0.5$ from open-loop prediction in task i)

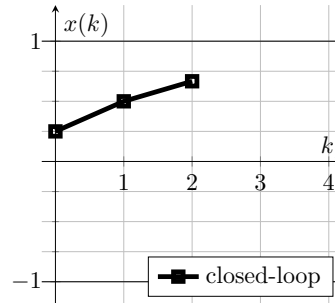
Notice that minimal possible input ($u = -1/3$) needs to be applied, in order to satisfy the constraints of the prediction horizon.

$$x_0 = 0.5$$

$$x_1 = 2x_0 - 1/3 = 2/3$$

$$x_2 = 2x_1 - 1/3 = 4/3 - 1/3 = 1$$

Plot $x(2) = 2/3$ in Figure b)



- iii) Notice $x(1)$ is on the boundary of the feasible set, since applying the minimal input twice is required to ensure $x_2 = 1$ in the prediction.

Lower point equal to -1 because positive inputs not constrained.

$$\mathcal{X}_N = \left\{ x \mid \begin{bmatrix} 1 \\ -1 \end{bmatrix} x \leq \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} \right\}$$

- iv) No, applying the minimal possible input will still result in state constraint violations for the second prediction step.

- v) Lower bound stays the same as for MPC without terminal constraint.

To find upper bound, apply max input twice.

$$x_2 = 0$$

$$x_2 = 2x_1 - \frac{1}{3} = 2 \left(2x_0 - \frac{1}{3} \right) - \frac{1}{3} = 4x_0 - \frac{3}{3} = 0$$

$$\Rightarrow x_0 = \frac{1}{4}$$

$$\mathcal{X}_N = \left\{ x \left| \begin{bmatrix} 1 \\ -1 \end{bmatrix} x \leq \begin{bmatrix} 0.25 \\ 1 \end{bmatrix} \right. \right\}$$

vi) Yes. Given the MPC problem is feasible at time step 0, we have the optimal state sequence $\{x_0^*, x_1^*, x_2^* = 0\}$ and input sequence $\{u_0^*, u_1^*\}$.

Applying u_0^* we reach the state x_1^* , where we know the feasible state sequence $\{x_1^*, 0, 0\}$ and input sequence $\{u_1^*, 0\}$, since from the dynamics $2 * 0 + 0 = 0$.

vii) False (the feasible set is decreased, in order to prevent state constraint violations at the third step),
True (for each $x(k)$ with feasible state/input sequence for (2) with $N = 2$, appending 0 is feasible for (2) with $N = 3$,

False ((2) has an additional constraint, therefore it has greater or equal optimal cost compared to (1) for the same (feasible) initial condition),

False (intuitively, with a longer horizon we have more "time" to reach 0, which requires smaller inputs and thus leads to a smaller cost; specific counterexample: $N_1 = 2$, $N_2 = 1$, $x_0 = -1$, for which $J_{N_1,2}^* < J_{N_2,2}^*$).

Exercise 3 (Exam 2018) [3pt. | 4pt.]

Consider the following autonomous system with an orthogonal dynamics matrix

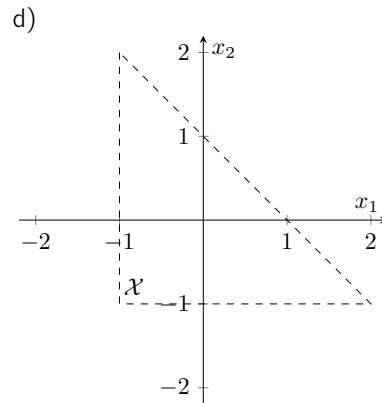
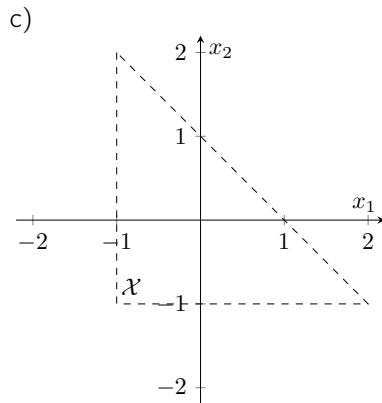
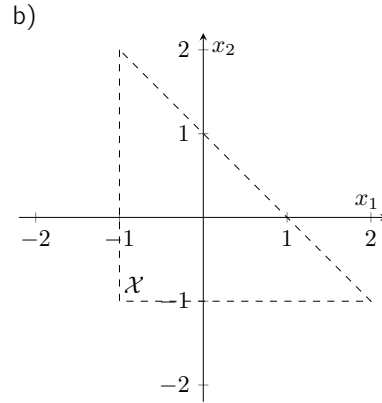
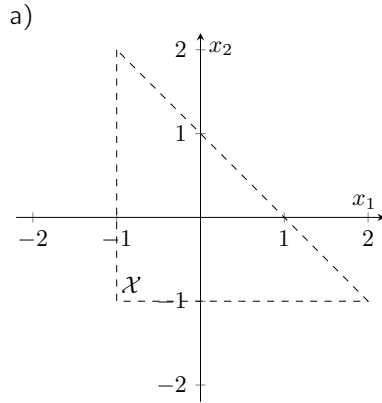
$$x(k+1) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} x(k) \quad (3)$$

and state constraints $\mathcal{X}$ as defined in the plots provided below, shown as dashed line.

Hint: For orthogonal matrices it holds that $A^T A = I$.

i) Draw the pre-set $\text{pre}(\mathcal{X})$, as well as $\text{pre}(\mathcal{X}) \cap \mathcal{X}$ in Figure a).

ii) Graphically find the maximum positive invariant set for system (3) under constraints $\mathcal{X}$, and draw it in Figure d). You can use the remaining plots b) - c) for intermediate steps.

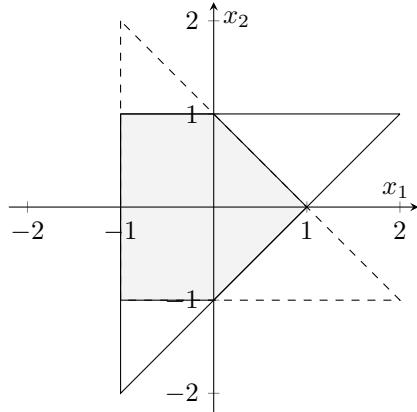


Solution:

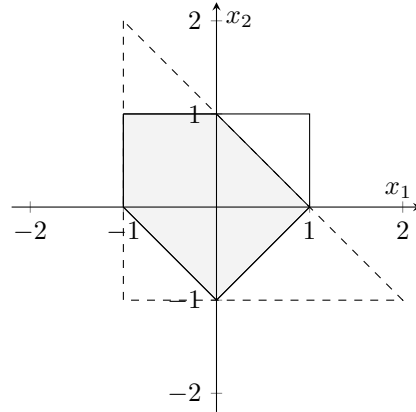
- i) pre set: $\{x | Ax \in \mathcal{X}\} = \{A^{-1}x | x \in \mathcal{X}\}$, $A^{-1} = A^T$ (1pt)
 Sketch $Pre(\mathcal{X})$
 Sketch intersection

- ii)

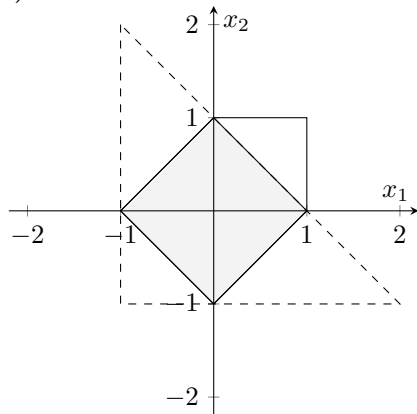
a)



b)



c)



d)

